PROGRAMMING FUNDAMENTALS

LABORATORY ASSIGNMENT — 03

Submitted By: Hamna Manzoor

Section: D

Department: BS Computer Science

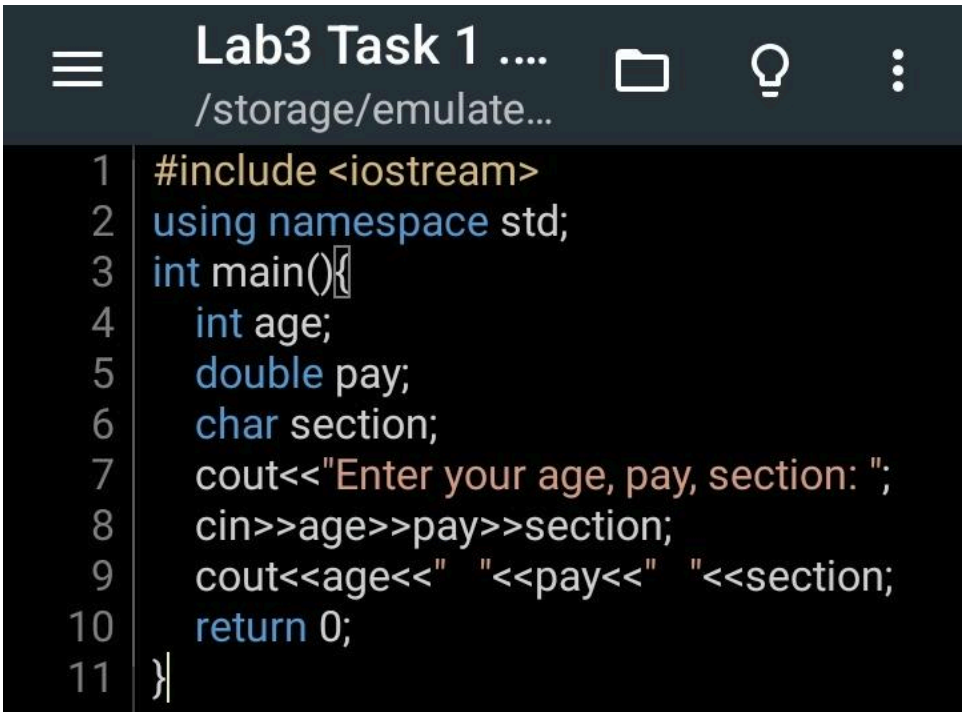
Subject: Programming Fundamentals

Lab: PF Lab 3

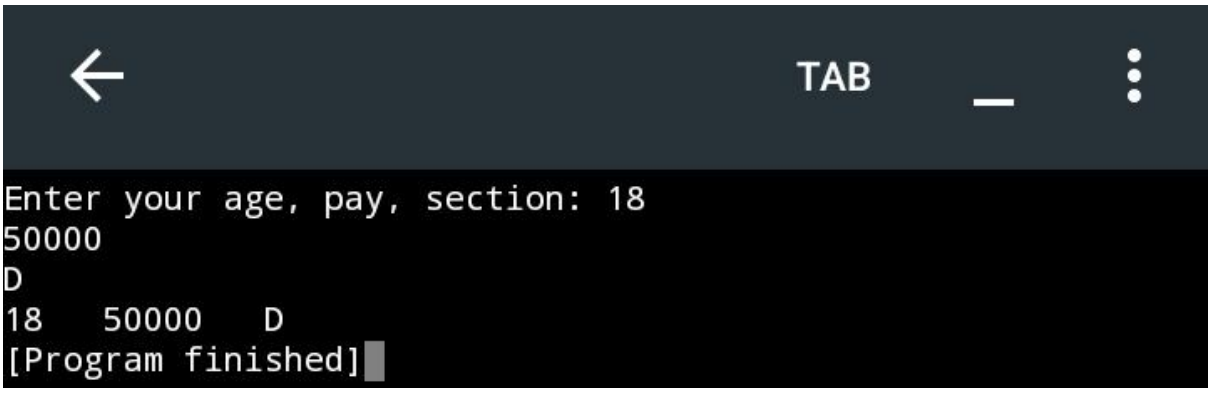
Task – 01: (cin Statement)

Assume variables int age; double pay; char section; Write single cin statement that will read input into each of these variables.

Program:



Output:



TASK-1 (cin Statement)

Assume variables int age; double pay; char section; Write single cin statement that will read input into each of these variables.

Program:

```
#include <iostream>
using namespace std;
int main(){
    int age;
    double pay;
    char section;
    cout << "Enter your age, pay, section: ";
    cin >> age >> pay >> section;
    return 0;
    }
```

Output

```
Enter your age, pay, section: 18
50000
D
18    50000    D
```

Task – 02 (Complete table)

Find Values:

28/4-2	5
6+12*2-8	22
4+8*2	20
6+17%3-2	6
2+22*(9-7)	46
(8+7)*2	30
16+7%2-1	0
12/(10-6)	3
(19-3)*(2+2)/4	16

TASK - 2 (Find Values)

28/4 - 2	5
6 + 12 * 2 - 8	22
4 + 8 * 2	20
6 + 17 % 3 - 2	6
2 + 22 * (9 - 7)	46
(8 + 7) * 2	30
16 + 7 % 2 - 1	0
12 / (10 - 6)	3
(19 - 3) * (2 + 2) / 4	16

Task – 03 (Automatic Type Conversion)

Assume the program have the following variable definitions:
int units;
float mass;
double weight;
weight = mass * units;
Which automatic data type conversion will take place?

A. mass is demoted to an int, units remains an int, and the result of mass * units is an int.
B. units is promoted to a float, mass remains a float, and the result of mass * units is a float.
C. units is promoted to a float, mass remains a float, and the result of mass * units is double.

In mass _ units, units is int so it is promoted to float. mass remains float and the result of mass _ units is float.

Then in weight = mass _ units, that float result is promoted to double to store in weight. So if we consider the whole statement, the final value stored in weight is double which is option C.

So for mass _ units answer is B, for weight = mass * units answer is C.

TASK-3 (Type Conversion)

Assume the program have the following variable defination:

```
int units;
float mass;
double weight;
weight = mass * units;
```

Which automatic data type conversion will take place?

Answer

In mass * units, units is int. units is int so it is promoted to float. Mass remains float and the result of mass * unit is float. Then, in weight = mass * units, that float result is promoted to double to store in weight. So if we consider whole statement, the final value stored in weight is double which is option C.

Task – 04 (Narrowing Conversion)

Assume a program has the following variable definitions:

int a, b = 2; float c = 4.2;

and the following statement: a = b * c;

What value will be stored in a?

A. 8.4
B. 8
C. 0
D. None of the above

Answer is B (8).

Reason: b _ c = 2 _ 4.2 = 8.4. 8.4 is float. When float is stored in int variable a, decimal part is truncated, not rounded. So a becomes 8.

TASK - 4 (Conversion)

Assume a program has following variable defination:

int a, b = 2; float c = 4.2;

and the following statement: a = b*c;

What value will be stored in a?

A. 8.4

B. 8

C. 0

D. None of these.

Answer: 8 will be stored in a.

Reason: b*c = 2*4.2 = 8.4.

8.4 is float. When float is stored in int variable a, decimal part is truncated, not rounded. So, a becomes 8.

Task – 05 (Type Casting)

Assume that qty and salesReps are both integers. Use a type cast expression to rewrite the following statement so it will no longer perform integer division.

unitsEach = qty / salesReps;

Answer: unitsEach = (float) qty / salesReps;

Reason: qty and salesReps both are int. So, qty / salesReps does integer division and decimal part is lost. If we cast one operand to float, it becomes float division and gives correct decimal result.

TASK - 5 (Type Casting)

Assume that qty and salesReps are both integers. Use a type cast expression to rewrite the following statement so it will no longer perform integer division.

```
unitsEach = qty / SalesReps;
```

Answer:

```
unitsEach = (float) qty / salesReps;
```

Reason: qty and SalesReps are both integers. So, qty / SalesReps does integer division and decimal part is lost. If we cast one operand to float, it becomes float division and gives correct decimal result.

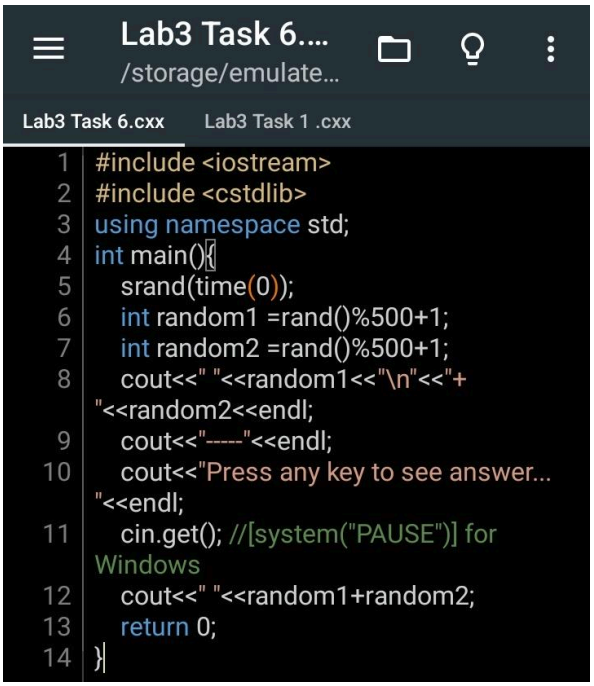
Task – 06 (Write A Program)

Write a program that can be used as a math tutor for a young student. The program should display two random numbers to be added, such as

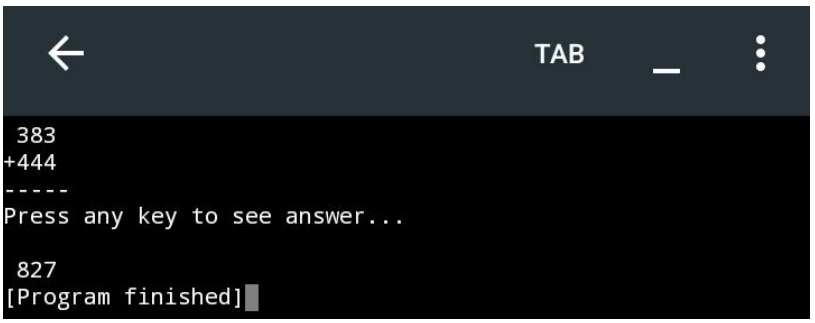
```
  247
+ 129
```

The program should then pause while the student works on the problem. When the student is ready to check the answer, he or she can press a key and the program will display the correct solution.

Program:



Output:



TASK- 6 (Write A Program)

Program:

```
#include <iostream>
#include <cstdlib>
using namespace std;
int main(){
srand(time(0));
int random1 = rand()% 500+1;
int random2= rand()% 500+1;
cout << "  " << random1 << "\n" <<"+"<<random2;
cout << "\n------" << endl;
cout << " Press any key to see answer.... "<<endl;
cin.get(); //[system("PAUSE")] for windows.
cout << "  " << random1 + random2;
   return 0;
}
```

Output

```
  183
+ 444
------
Press any key to see answer....

  827
```

Home Task – 01 (Error Finding / Debugging)

Each of the following programs has some errors. Locate as many as you can.

Program 1:

```
1  using namespace std;
2  void main ()
3  {
4      double number1, number2, sum;
5      cout << "Enter a number: ";
6      Cin << number1;
7      cout << "Enter another number: ";
8      cin << number2;
9      number1 + number2 = sum;
10     cout "The sum of the two numbers is "
   << sum
11 }
```

Full list of errors in Program-1:

1. `#include <iostream>` is missing

2. `void main()` should be `int main()`

3. `Cin << number1;` -> `cin >> number1;` - C is capital and operator wrong

4. `cin << number2;` -> should be `cin >> number2;`

5. `number1 + number2 = sum;` -> should be `sum = number1 + number2;`

6. `cout "The sum...` -> missing `<<` after cout

7. `... << sum` -> missing `;` at end

8. missing `return 0;`

HOME TASK - 01 (Find Errors)

Program 1:

```
using namespace std;
Void main()
{
  double number1, number2, sum;
  Cout << "Enter a number:";
  Cin << number1;
  Cout << "Enter another number: ";
  Cin << number2;
  number 1 + number 2 = sum;
  Cout "The sum of the two numbers is "
<< sum
}
```

Errors:

1. '# include <iostream>' is missing.
2. 'void main()' should be 'int main()'.
3. 'Cin << number1;' should be 'cin >> number1;'.
4. 'cin << number2;' should be 'cin >> number2;'.
5. 'number1 + number2 = sum;' should be 'sum = number1 + number2;'
6. 'Cout "The sum ....." missing' << ' after Cout.
7. "<< sum" should be "<< sum;".
8. missing 'return 0;'.

Program 2:

```
1  #include <iostream>
2  using namespace std;
3  void main()
4  {
5      int number1, number2;
6      float quotient;
7      cout << "Enter two numbers and I will
   divide\n";
8      cout << "the first by the second for you.
   \n";
9      cin >> number1, number2;
10     quotient = float<static_cast>(number1)
   / number2;
11     cout << quotient
12 }
```

Full list of errors in Program-2:

1. `void main()` should be `int main()`

2. `cin >> number1, number2;` should be `cin >> number1 >> number2;`

3. `float<static_cast>(number1)` should be `static_cast<float>(number1)`

4. `cout << quotient` missing `;`

5. `return 0;` missing

HOME TASK - 01 (Find Errors)

Program 1:

```
#include <iostream>
using namespace std;
void main()
{
  int number1, number2;
  float quotient;
  cout << "Enter two numbers and I will divide \n";
  cin >> number1, number2;
  quotient = float < static_cast > (number1)
  / number2;
    cout << quotient
  }
```

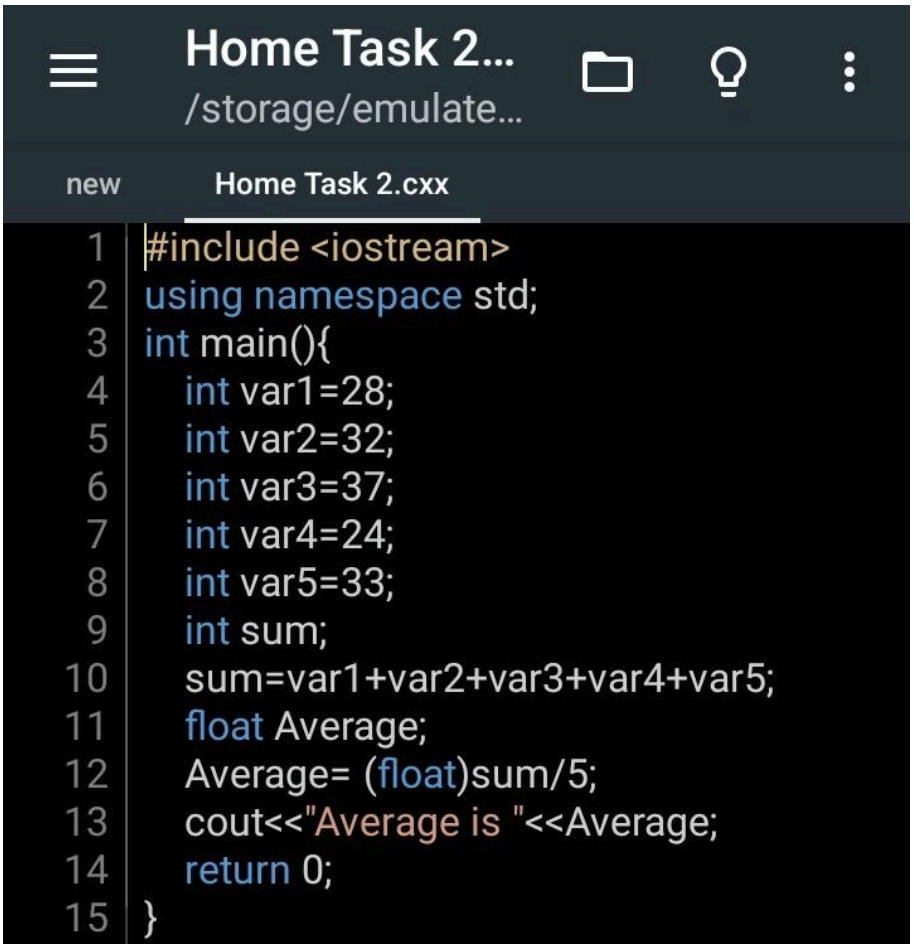
Errors:

1. 'void main' should be 'int main'.
2. 'cin >> number1 , number2;' should be 'cin>>number1>>number2;'.
3. 'float <static_cast>' should be ' static_cast <float>'.
4. 'cout << quotient' missing ';'.
5. 'return 0;' missing.

Home Task – 02 (Average of Values Program)

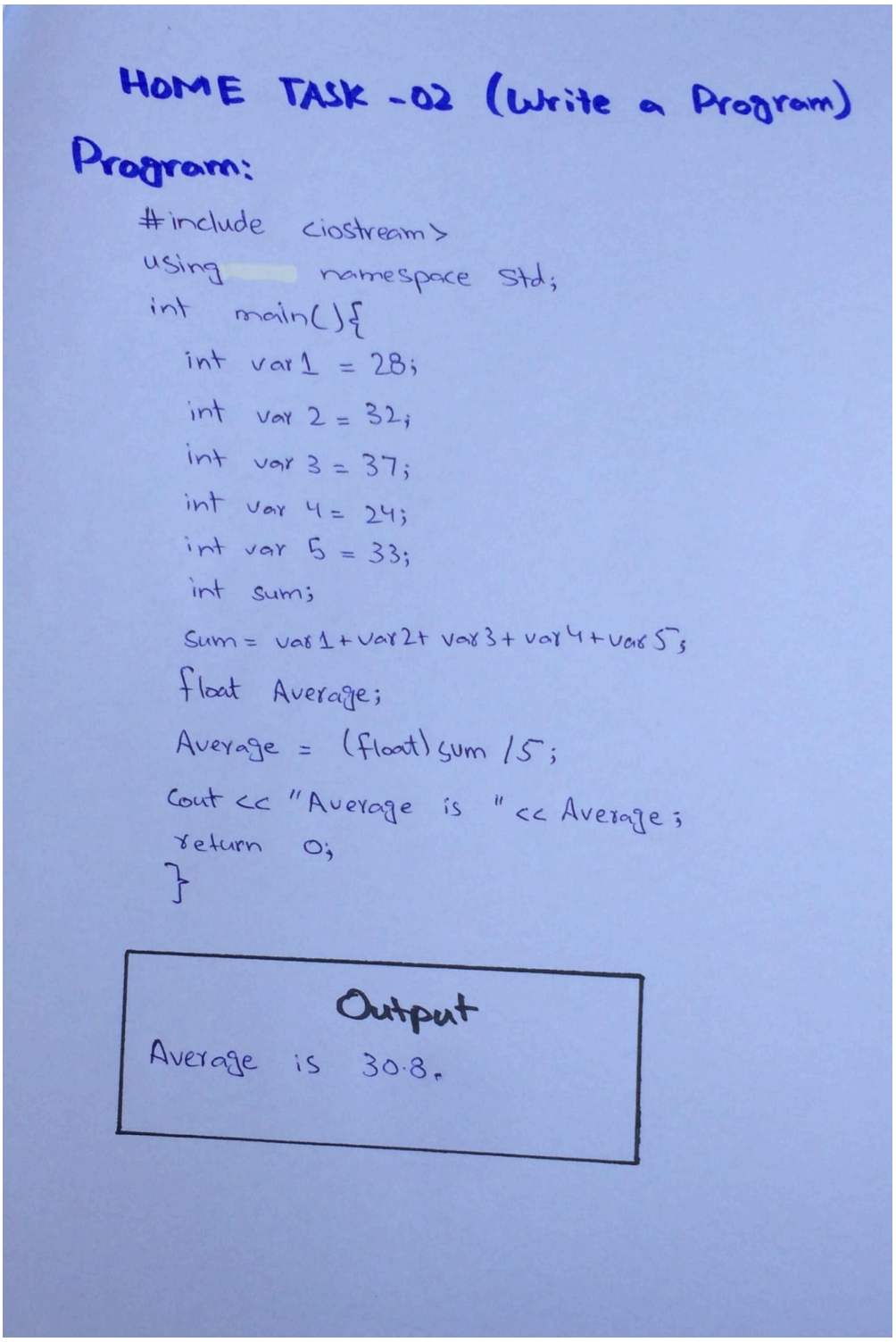
Average of Values to get the average of a series of values, you add the values up and then divide the sum by the number of values. Write a program that stores the following values in five different variables: 28, 32, 37, 24, and 33. The program should first calculate the sum of these five variables and store the result in a separate variable named sum. Then, the program should divide the sum variable by 5 to get the average. Display the average on the screen.

Program:



Output:





HOME TASK -02 (Write a Program)

Program:

```
#include <iostream>
using namespace std;
int main(){
  int var1 = 28;
  int var 2 = 32;
  int var 3 = 37;
  int var 4 = 24;
  int var 5 = 33;
  int sum;
  Sum = var1+var2+ var3+ var4+var5;
  float Average;
  Average = (float) sum /5;
 Cout << "Average is " << Average;
 return 0;
}
```

Output

Average is 30.8.